

Partial-Homogeneity-Based Two-Dimensional High-Resolution Nuclear Magnetic Resonance Spectroscopy under Inhomogeneous Magnetic Fields

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High-resolution multidimensional nuclear magnetic resonance (NMR) spectroscopy serves as an irreplaceable and versatile tool in various chemical investigations. In this study, a method based on the concept of partial homogeneity is developed to offer two-dimensional (2D) high-resolution NMR spectra under inhomogeneous fields. Oscillating gradients are exerted to encode the high-resolution information, and a field-inhomoge-

neity correction algorithm based on pattern recognition is designed to recover high-resolution spectra. Under fields where inhomogeneity primarily distributes along a single orientation, the proposed method will improve performances of 2D NMR spectroscopy without increasing the experimental duration or significant loss in sensitivity, and thus may open important perspectives for studies of inhomogeneous chemical systems.

1. Introduction

Multidimensional (mD) nuclear magnetic resonance (NMR) spectroscopy is considered to be a versatile and powerful method for characterizing the structures and dynamics of molecules. A lot of progress in mD NMR spectra has been witnessed in the past few decades.^[1] However, there are occasions for which magnetic fields that are sufficiently homogeneous are simply unavailable, even with advanced shimming techniques,^[2] for example, in spatially heterogeneous chemical systems or remote NMR characterizations. As a result, adjacent peaks in mD NMR spectra may overlap each other, thus deteriorating spectral readability and resulting in loss of information. Various methods have been presented to help overcome this problem, for instance, zero-quantum coherence (ZQC),^[3] spin echo correlated spectroscopy (SECSY),^[4] the intermolecular multiquantum coherence (iMQC) technique,^[5] and schemes developed from the spatiotemporal encoding technique.^[6] Moreover, with spatial distributions of field inhomogeneity mapped, special radiofrequency pulses can be designed to compensate for field inhomogeneity.^[7] In addition, magic-angle spinning, which provides high-resolution spectra even under inhomoge-

neous fields, constitutes a widely adopted technique for the study of solid-state chemical systems.^[8]

Recently, pattern recognition has been applied with partial homogeneity (PHASICS)^[9] and line-broadening interference (LBI)^[10] to yield high-resolution 1D NMR spectra under inhomogeneous fields. In this study, PHASICS is equipped with different evolution and mixing periods, which can provide one free evolution in progressive delays (t_1), to offer different types of 2D NMR information in unison with a designed 2D field-inhomogeneity correction algorithm. Theoretical deductions along with experiments have been carried out to verify the feasibility and effectiveness of PHASICS in yielding high-resolution 2D NMR spectra under inhomogeneous fields.

2. Theory

Schematic diagrams of pulse sequences to yield 2D PHASICS spectra are shown in Figure 1. Different mixing periods can be integrated to construct different 2D NMR spectroscopy. The concept of partial homogeneity originates from the simple fact that sufficiently small voxels are intrinsically associated with high resolutions. The additional mixing period, along with the ensuing gradients (G_{sx} , G_{sy} and G_{sz}), is thus irrelevant to obtaining high resolution and is neglected in the following deductions. The effects of molecular diffusion and relaxation are ignored, and all RF pulses are applied along the x direction. Without loss of generality, an AX spin-1/2 system is considered. Spins I and S characterize chemical shifts of Ω_I and Ω_S with a coupling constant J . The spatial distribution of magnetic-field inhomogeneity is denoted as $\Delta B(r)$, where r refers to x , y , and z . For descriptive convenience, some symbols are defined as follows: T_g denotes the duration of each single gradient lobe; t_a the time running from the beginning of the positive or negative gradient lobe to a certain acquisition moment within that

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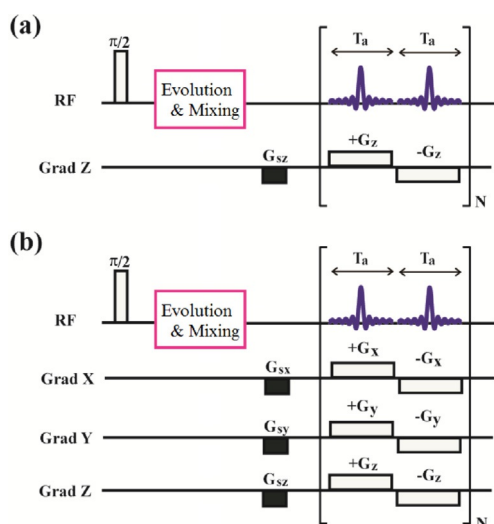


Figure 1. Pulse sequences for high-resolution 2D spectra: a) Z-PHASICS and b) R-PHASICS. N indicates the number of pairs of oscillating gradients; T_a denotes the duration of each single gradient lobe; and G_{sx} , G_{sy} , and G_{sz} indicate shifting gradients along the X , Y , and Z axes, which are employed to shift the echo center.

gradient lobe; N the number of pairs of oscillating gradients; n a running variable of $n = 1, 2, \dots, N$; and G_x , G_y , and G_z are the intensities of the oscillating gradients during the acquisition period along the X , Y , and Z axes, respectively. Regarding the spin system discussed herein, the reduced density operators at the initial thermal equilibrium state with a high-temperature approximation are given as Equation (1):

$$\sigma_{\text{eq}} = I_z + S_z \quad (1)$$

where the Boltzmann factor is omitted for clarity; I_z and S_z represent longitudinal components of I and S spins, respectively. The two sequences shown in Figure 1 differ from each other in the way the encoding gradients are implemented. Figure 1a utilizes encoding gradients along the Z axis merely (denoted as Z-PHASICS) and is applicable to cases where the field inhomogeneity along the Z axis dominates. For a more general case, where field inhomogeneity distributes along an axis that is different from any of the coordinate X , Y , and Z axes (denoted as the R axis), R-PHASICS, which simultaneously employs X , Y , and Z gradients to generate an effective gradient field along the R axis, in Figure 1b, can be used.

We first consider evolutions of I spins in Z-PHASICS. As mentioned before, the Z-PHASICS deals with cases where field inhomogeneity along the Z axis, namely, $\gamma\Delta B(r) \approx \gamma\Delta B(z)$. Regarding a certain acquisition moment within the $(2n-1)$ lobe of gradient in the acquisition period, the density operator corresponding to detectable signals evolves to give Equation (2):

$$\sigma'_{\text{obs}} = \frac{I^-}{2i} e^{i[\Omega_I + \gamma\Delta B(z)][(2n-2)T_a + t_a]} \cos\{\pi J[(2n-2)T_a + t_a]\} e^{i\gamma G_z z t_a} \quad (2)$$

As evolutions of S spins are similar to those of I spins, the density operator of S spins evolves to be Equation (3):

$$\sigma_{\text{obs}}^S = \frac{S^-}{2i} e^{i[\Omega_S + \gamma\Delta B(z)][(2n-2)T_a + t_a]} \cos\{\pi J[(2n-2)T_a + t_a]\} e^{i\gamma G_z z t_a} \quad (3)$$

Therefore, on the assumption that M_0 is the equilibrium magnetization, the detected signal can be written as Equation (4):

$$\begin{aligned} M_{\text{obs}}(n, t_a) &\propto M_0 \int_r \text{Tr}[(\sigma'_{\text{obs}} + \sigma_{\text{obs}}^S)(I^+ + S^+)] dr \\ &\propto M_0 \int_r \sum_{\xi=L,S} e^{i[\Omega_{\xi} \pm \pi J + \gamma\Delta B(z)](2n-2)T_a} e^{i[\gamma G_z z + \Omega_{\xi} \pm \pi J + \gamma\Delta B(z)]t_a} dr \end{aligned} \quad (4)$$

Two variables (n and t_a) are available for performing 2D Fourier transformation (FT). After 2D FT to the theoretical detectable signal, we can obtain the spectral information of [Eq. (5)]:

$$\begin{aligned} S(\omega_2, \omega_3) &\propto M_0 \int_r \sum_{\xi=L,S} \delta\{\omega_2 - [\Omega_{\xi} \pm \pi J + \gamma\Delta B(z)]\} \\ &\quad \times \delta\{\omega_3 - [\gamma G_z z + \Omega_{\xi} \pm \pi J + \gamma\Delta B(z)]\} dr \end{aligned} \quad (5)$$

where δ denotes the Dirac delta function. As the dispersive frequency range by encoding gradients along the Z axis is far larger than the width of chemical shifts, scalar coupling constants, or line broadening by magnetic-field inhomogeneity, we can rewrite Equation (5) as Equation (6):

$$S(\omega_2, \omega_3) \propto M_0 \int_r \sum_{\xi=L,S} \delta\{\omega_2 - [\Omega_{\xi} \pm \pi J + \gamma\Delta B(z)]\} \delta\{\omega_3 - \gamma G_z z\} dr \quad (6)$$

From Equation (6), it can be seen that field inhomogeneity serves as frequency offsets to shift chemical sites in different layers (different z). Within each different layer, high-resolution spectral information is maintained.

Theoretical treatment of Z-PHASICS is also valid for R-PHASICS by replacing Z with R in Equations (2)–(6). Note that prior knowledge about field inhomogeneity is necessary to facilitate the parameter settings of the X , Y , and Z encoding gradients. The effective encoding gradient should be applied along the R axis, where field inhomogeneity is distributed. Similar to Z-PHASICS, R-PHASICS is capable of compensating high-order field inhomogeneity along this R axis.

The mixing period along with the evolution period, which is achieved by adding one free evolution in progressive delays (t_1), can be directly combined with Z-PHASICS or R-PHASICS to yield high-resolution 2D spectra. Although the resulting 2D high-resolution spectra are obtained through projections from 3D data (namely t_1 , n , and t_a), the time cost does not increase, because echo planar spectroscopic imaging (EPSI) module^[11] is employed to accelerate the encoding process.

The maintained high-resolution information, as illustrated by Equation (6), can be recovered with the aid of pattern recognition. The 2D inhomogeneity correction algorithm is designed and implemented step by step as follows: 1) reconstruct the

3D data $D(t_1, n, t_a)$, perform 3D FT to obtain the frequency-domain information $S(\omega_1, \omega_2, \omega_3)$ (ω_1 , ω_2 , and ω_3 represent the F1, F2, and F3 dimensions, respectively) and break the 3D data into a series of 2D data [denoted as $M(s)$] along the ω_3 dimension; 2) binarize $M(s)$ to offer $I(s)$, apply dilation operator to $I(s)$, and then extract the contours; 3) average contour coordinates of different peaks independently to yield central coordinates of different peaks in $I(s)$ and compare central coordinates among $I(s)$ to provide correction information; and 4) perform circular shifting on $S(\omega_1, \omega_2, \omega_3)$ along both ω_1 and ω_2 dimensions according to the correction information and supply high-resolution information by projecting $S(\omega_1, \omega_2, \omega_3)$ onto the $\omega_1\omega_2$ plane.

To investigate the signal-to-noise (SNR) performance of PHASICS, a comparison between PHASICS and conventional single-pulse spectroscopy (denoted as CONV) is performed. As CONV will suffer from severe SNR loss under inhomogeneous fields, the comparison is drawn under the homogeneous field. In PHASICS experiments, the receiver bandwidths are enlarged to cover the dispersive frequency range induced by encoding gradients. Theoretically, the enlargement of the receiver bandwidth will introduce more noise to degenerate the SNR. Note that the digital filters are generally available in modern NMR spectrometers to ameliorate this sensitivity loss. On the other hand, the oscillating gradients at high slew rates will attenuate the SNR performance by way of diffusion. As a result, CONV characterizes SNR advantage over PHASICS under homogeneous fields. For inhomogeneous fields, both CONV and PHASICS will undergo SNR loss. The SNR downtrend of CONV is more severe than that of PHASICS, due to the sensitivity to magnetic-field inhomogeneity.

3. Results and Discussion

Figure 2 presents the pilot course of the 2D field-inhomogeneity correction algorithm. The shimming coils from Z1 to Z7 are detuned simultaneously (denoted as Z1~Z7 inhomogeneity). The experimental data from Z-PHASICS-COSY (homonuclear correlation spectroscopy), which is integrated with the module of $t_1 \sim 90^\circ$, is used to illustrate the procedures step by step. It can be seen from Figure 2a that the spectral lines behave as curved lines, due to the presence of the inhomogeneity. First, the original Z-PHASICS-COSY spectrum is dismantled into a series of high-resolution 2D COSY spectra corresponding to different Z (Figure 2d). The 2D COSY spectra are then binarized and dilated to provide images (Figure 2c) with holes that are located where the spectral peaks occur. After the contour extraction operation, contours of spectral peaks are obtained as shown in Figure 2e. Thereafter, the central coordinates of peaks are calculated by averaging the contour coordinates to provide correction information (Figure 2f). The circular shifting operations are performed on the series of COSY spectra (Figure 2d) along both the F1 and F2 dimensions according to the correction information. Finally, spectral lines in the corrected Z-PHASICS-COSY spectrum (Figure 2b) extend as straight lines in parallel to the Z axis, signifying that the high-resolution spectrum can be retrieved through projection onto the F1–F2 plane after corrections.

Figure 3 conveys the results of conventional techniques and Z-PHASICS method under the Z1–Z7 inhomogeneity. From Figures 3a and c, the Z1~Z7 inhomogeneity greatly distorts the lineshape rather than sharply broadening the line width. As a result, the *J*-resolved and COSY spectra exhibit appear more

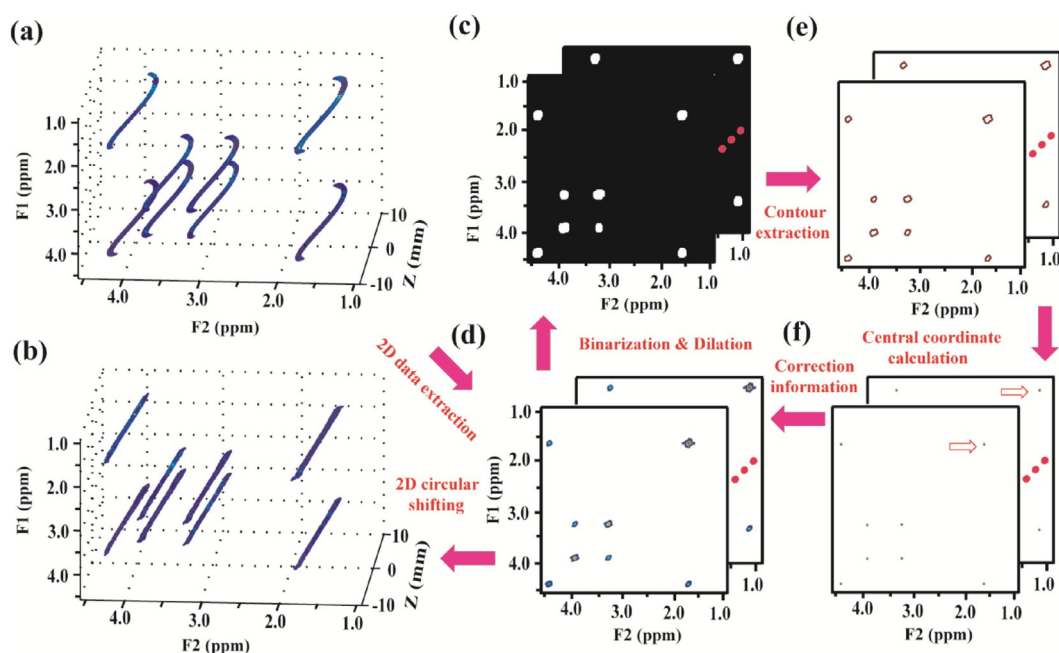


Figure 2. Pilot process of the 2D field inhomogeneity correction algorithm. a) The original Z-PHASICS-COSY spectrum. d) A series of 2D data that was extracted from (a). c) Images obtained after performing binarization and dilation of (d). e) The extracted contour of the peaks. f) The central coordinates of peaks calculated from the corresponding contours. b) The recovered high-resolution Z-PHASICS-COSY spectrum after 2D circular shifting to (d) with the aid of correction information provided by (f).

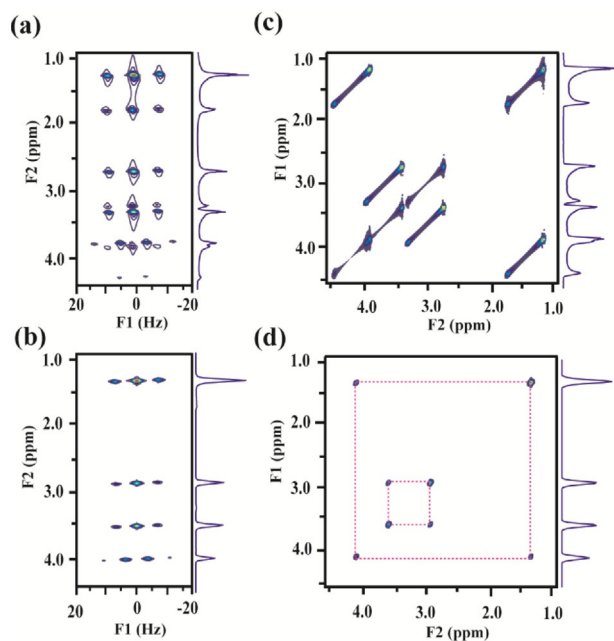


Figure 3. Spectra obtained by a) conventional COSY and *J*-resolved spectroscopy under the same inhomogeneous field, b) Z-PHASICS-*J*, c) COSY, and d) Z-PHASICS-COSY along with projections. The red dotted box in (d) shows the coupling structures.

complicated, thus resulting in misleading spectral interpretation and making assignments difficult. In contrast, the Z-PHASICS-*J* and Z-PHASICS-COSY spectra exhibit the correct scalar-coupling information and coupling structures. The resolution improvements can also be observed by comparing the 1D projection (see the Supporting Information for the pilot correction process). Therefore, the Z-PHASICS-*J* and Z-PHASICS-COSY are capable of yielding high-resolution information under single axially inhomogeneous fields with high-order terms.

A quinine solution is used to investigate the performance of Z-PHASICS when dealing with more complicated chemical systems (Figure 4). The shimming coils from Z1 to Z7 are detuned simultaneously. Modules of $t_1/2 \sim 90^\circ \sim t_1/2$ and $t_1/2 \sim 180^\circ \sim t_1/2$ are integrated with Z-PHASICS to construct Z-PHASICS-SECSY and Z-PHASICS-*J* spectra. The reason for placing $t_1/2 \sim 90^\circ \sim t_1/2$ instead of $t_1 \sim 90^\circ$ lies in the fact that the dephasing induced by the field inhomogeneity during the evolution period is sufficiently large to influence G_{SE} , which is used to shift the echo centers for optimal sensitivity and resolution. Therefore, the spin echo correlated module is employed to refocus dephasing during the evolution period. Figure 4b exhibits scalar-coupling networks clearly and Figure 4c presents delicate scalar-coupling splitting patterns under severe inhomogeneity.

Butyl methacrylate was used in the heteronuclear single quantum correlation (HSQC) experiment to test the per-

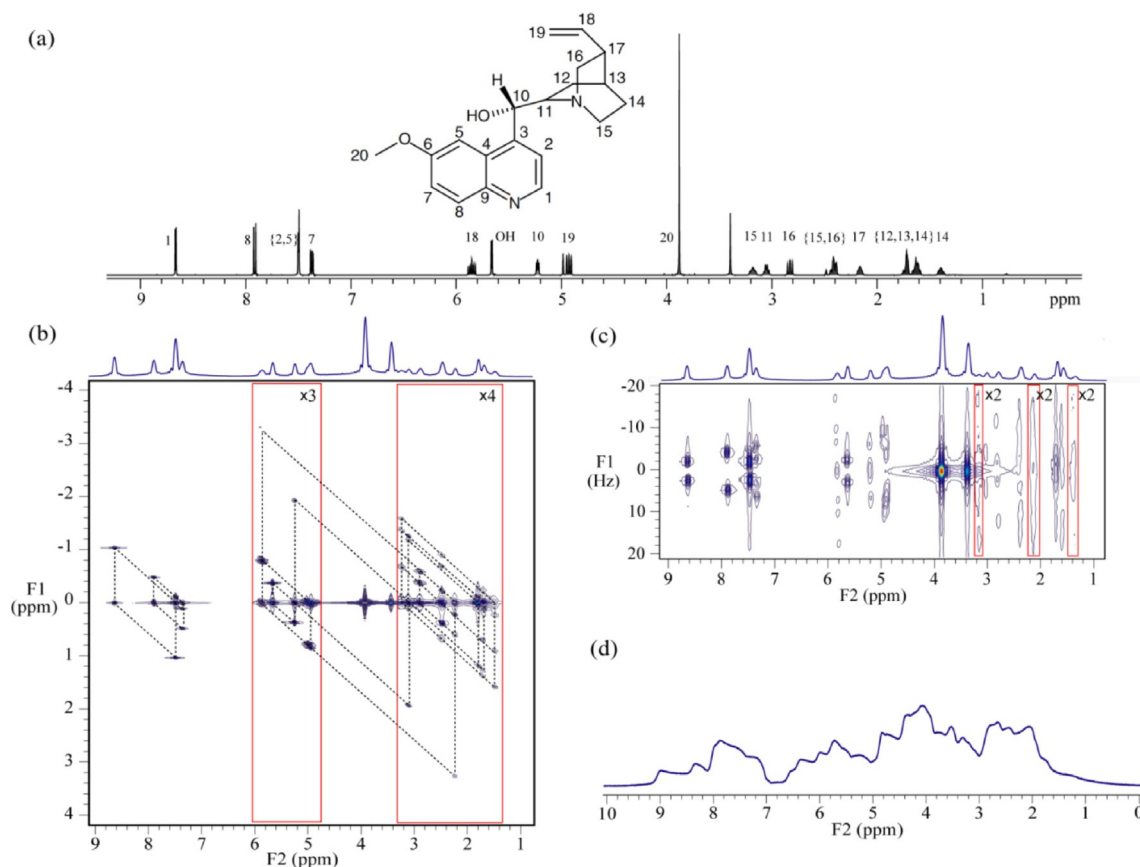


Figure 4. a) The molecular structure and standard 1D NMR spectrum of quinine, b, c) Z-PHASICS-*J* and Z-PHASICS-SECSY spectra recorded under an inhomogeneous field introduced by detuning shimming coils from Z1 to Z7, and d) the 1D spectrum under the same inhomogeneous field.

formance of Z-PHASICS in heteronuclear experiments under inhomogeneous fields. The shimming coils from Z1 to Z7 are detuned simultaneously. The sequence used in this experiment is illustrated in Figure 5a. Accurate correlation information between ^1H and ^{13}C shifts can be observed from Z-PHASICS-HSQC (Figure 5b), whereas the over-low resolution in the F2 dimension of the conventional HSQC renders ^1H chemical shifts difficult to distinguish. Peaks in the Z-PHASICS-HSQC spectrum suffer from heavy-tailed noise; the reason for this lies in the dephasing introduced by field inhomogeneity during the evolution period. The dephasing leads to shifting of echo centers, and the G_{sz} set previously may no longer be suitable with scans. The Z-PHASICS-HSQC spectrum still exhibits higher resolution and provides information that is much more reliable than conventional HSQC spectra does under the same inhomogeneous field. Notably, a conventional COSY experiment is characterized by identical time cost as that of its Z-PHASICS or R-PHASICS counterpart. So is the case for the J -resolved experiment and the corresponding Z-PHASICS or R-PHASICS version. In other words, the improvements in performance under 1D inhomogeneous fields are obtained without increasing the time cost.

The results of R-PHASICS experiments together with that of their conventional counterparts are presented in Figure 6. The modules of $t_1/2 \sim 180^\circ \sim t_1/2$ and $t_1/2 \sim 90^\circ \sim t_1/2$ are combined with R-PHASICS to form R-PHASICS- J and R-PHASICS-COSY. The COSY spectrum can be recovered from spin echo correlated spectrum through a simple and direct mathematical conversion.^[5b] The XYZ inhomogeneity is sufficiently large to wipe

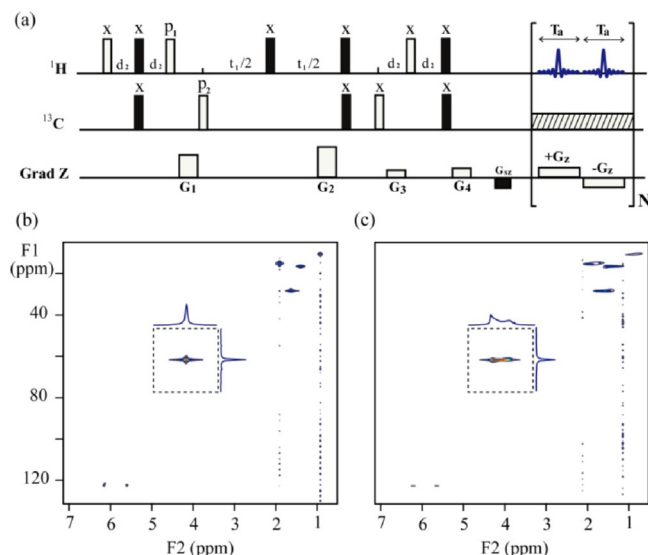


Figure 5. a) The sequence of Z-PHASICS-HSQC. The unfilled rectangles represent 90° pulses and the solid rectangles represent 180° pulses. N indicates the number of pairs of oscillating gradients; T_g denotes the duration of each single gradient lobe; and G_z denotes the shifting gradients along the Z axis. A four-step phase cycling was employed to improve SNR and alleviate pulse imperfections. The phases of pulse p_1 were altered by reference to $(y, y, -y, -y)$, phases of pulse p_2 $(x, -x, x, -x)$, and phase of receiver $(y, -y, -y, y)$. $d_2 = 1/2 J_{\text{CH}}$. b) Z-PHASICS-HSQC spectrum obtained under the inhomogeneous field. c) The conventional gHSQC spectrum recorded under the same inhomogeneous field.

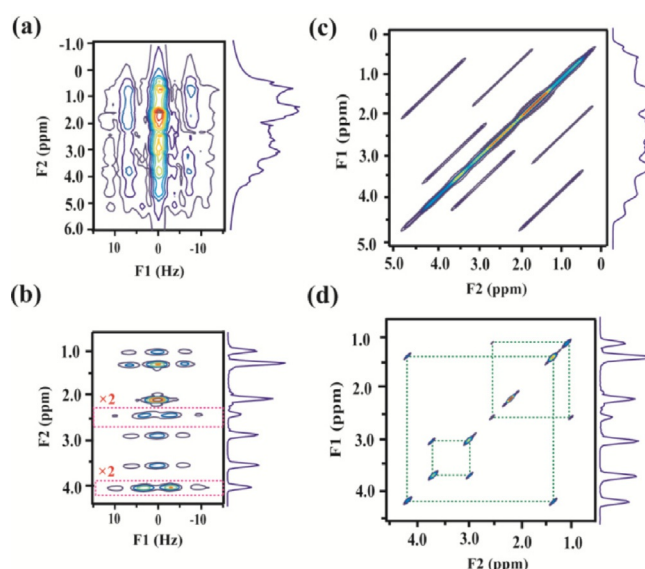


Figure 6. a) Conventional J -resolved spectrum, b) R-PHASICS- J spectrum, c) conventional COSY spectrum, and d) R-PHASICS-COSY spectrum obtained in the same inhomogeneous field with a line width of 1692.4 Hz. The peaks shown in the red dotted boxes in (b) are displayed with their intensities magnified by twofold. Green dotted boxes in (d) show the scalar-coupling networks.

out readable information in conventional J -resolved spectrum (Figure 6a). With the R-PHASICS- J , seven resonances located at 1.02 (triplet), 1.30 (triplet), 2.13 (singlet), 2.50 (quartet), 2.98 (triplet), 3.69 (triplet), and 4.19 ppm (quartet) were recovered (Figure 6b). Regarding the conventional COSY spectrum, diagonal peaks are broadened and overlap each other and cross peaks appear with broad linewidths (Figure 6c). Cross peaks located at (1.02, 2.50), (1.30, 4.19), and (2.98, 3.69) along with symmetrical ones were recovered with enhanced resolutions (Figure 6d; see the Supporting Information for pilot correction process).

When the field inhomogeneity and effective encoding gradient share an identical axis, R-PHASICS performs satisfactorily. However, the addition of high-order inhomogeneity tilts the direction of the field inhomogeneity away from the R axis, thus leading to performance degeneration. R-PHASICS experiments without mixing periods were performed under inhomogeneous fields with high-order terms to investigate this degeneracy (see the Supporting Information for details). R-PHASICS can perform satisfactorily in cases where the field inhomogeneity is primarily distributed along the R axis. The three simultaneously existing oscillating encoding gradients in the R-PHASICS sequence create an effective gradient-field along a direction that is different from any of the coordinate X , Y , and Z axes. Therefore, R-PHASICS is identical to Z-PHASICS except for the difference in the direction R , which is parallel to the direction of the effective gradient field, and effects induced by field inhomogeneity along this dimension can be removed. Note that actual 3D arbitrary field inhomogeneity may be too complicated to be modeled as inhomogeneity along a single direction, and thereafter both Z-PHASICS and R-PHASICS will suffer from performance degenerations under 3D field inhomogeneity. In dis-

positional systems with 3D field inhomogeneity, an alternative is to implement Z-PHASICS with two phase encodings^[9] along two additional directions. The experimental time would then be extended because of the two additional dimensions, calling for special designs to reduce the time cost.

To explore SNR performance, experiments utilizing Z-PHASICS under a series of inhomogeneous magnetic fields were carried out (Figure 7; see the Supporting Information for

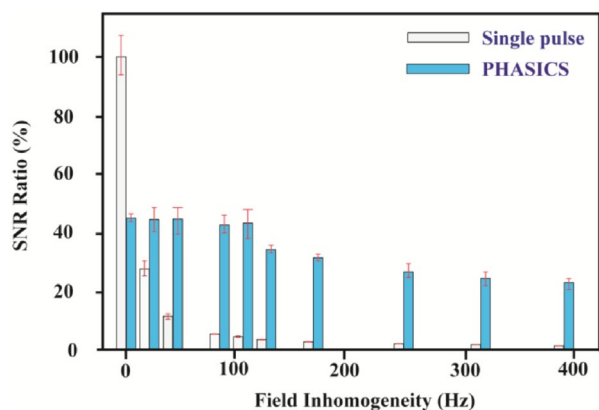


Figure 7. Quantitative comparisons of SNR performances between CONV and Z-PHASICS under a series of inhomogeneous fields with line width of 1.7, 23.7, 52.2, 81.3, 109.5, 133.3, 177.5, 257.4, 321.7, and 385.7 Hz, respectively.

more details). The SNR of conventional single-pulse spectroscopy declines dramatically with increasing field inhomogeneity. In contrast, Z-PHASICS exhibits comparatively stable SNR performance under varying fields. Although the Z-PHASICS spectrum is an accumulation of spectra, the diffusion effect may be induced by oscillating gradients at high slew rates. Therefore, the SNR of CONV is better than that of Z-PHASICS under homogeneous fields. In cases of inhomogeneous fields, the Z-PHASICS characterizes SNR advantages over CONV. The above quantitative analyses were executed on Z-PHASICS. However, the principle applies to other variations of the PHASICS method.

2D spectroscopy developed based on partial homogeneity presents better SNR and resolution performances under inhomogeneous fields, implying that partial-homogeneity-based 2D spectroscopy can serve as a helpful complement to its conventional counterparts under such circumstances.

The mD NMR spectroscopy can also be developed from ZQC or SECSY. The major problem lies in their incapability to generate chemical-shift information. In contrast, mD spectroscopy based on PHASICS provides chemical shifts directly. Applications of high-resolution mD spectroscopy schemes built on the basis of the iMQC technique are handicapped by expensive time costs, due to $(m+1)D$ acquisitions. The PHASICS method enhances the resolution without increasing the time cost. On the other hand, ZQC, SECSY, and iMQC based techniques perform better for 3D inhomogeneous fields. Methods developed from spatiotemporal encoding technique can be primarily categorized into three types: first, radiofrequency compensa-

tion^[6a,b] is combined by utilizing special prepared chirp pulses. Currently, it may be difficult to implement 3D compensations, and its performance may, thus be unsatisfactory under 3D inhomogeneity. Second, spin echo and J -detection modules^[6c] are integrated simultaneously, suffering from the same disadvantage as SECSY does (chemical-shift information is removed). Third, asymmetrical encoding/decoding^[6d] is used to execute encoding and decoding along three orthogonal dimensions in unison with asymmetrical gradients to overcome 3D first-order magnetic-field inhomogeneity. Its performances may not equal that of PHASICS in cases where field inhomogeneity is mainly single axial with high-order terms. Note that a favorable feature of the spatiotemporal encoding technique lies in its superior acquisition efficiency in comparison with all the other aforementioned techniques.

4. Conclusions

In this study, the concept of partial homogeneity was employed to generate high-resolution 2D NMR spectra along with a 2D field-inhomogeneity correction algorithm developed from pattern recognition. With the enhanced SNR and resolution performances under inhomogeneous magnetic fields, the PHASICS-based high-resolution 2D spectroscopy constitutes an alternative to conventional techniques. The proposed methods may open important perspectives for studies of inhomogeneous chemical systems.

Experimental Section

COSY and J -resolved spectroscopy were employed for comparisons with PHASICS under inhomogeneous fields. For convenience, some more symbols are defined: TR denotes the repetition time; ss the number of dummy scans to reach the steady-state of magnetization; SW_1 and SW_2 are spectral widths in the F1 and F2 dimensions, respectively and; ni is the number of points in the F1 dimension. $TR=4$ s and $ss=4$ were utilized in all experiments unless otherwise specified. The receiver bandwidths of PHASICS experiments were all 250 KHz.

Several experiments were performed to test the performance of Z-PHASICS. The shimming coils from Z1 to Z7 were detuned simultaneously.

First, modules of $t_1 \sim 90^\circ$ and $t_1/2 \sim 180^\circ \sim t_1/2$ were integrated with Z-PHASICS to construct Z-PHASICS-COSY and Z-PHASICS- J , respectively. An ethyl 3-bromopropionate sample was used. Experimental parameters that were shared by COSY and Z-PHASICS-COSY were: $SW_1=4.40$ ppm, $SW_2=4.40$ ppm, and $ni=256$ with corresponding total acquisition durations of about 17 min. The J -resolved and Z-PHASICS- J experiments were performed with: $SW_1=0.20$ ppm, $SW_2=4.40$ ppm, and $ni=50$ with total acquisition durations of about 4 min. The Z-PHASICS-COSY and Z-PHASICS- J experiments shared the following parameters: $T_a=224$ μ s, $N=200$, $G_z=29.30$ Gcm⁻¹, and $G_{sz}=-5.84$ Gcm⁻¹ with a duration of 0.45 ms.

Second, the Z-PHASICS-SECSY and Z-PHASICS- J experiments using the quinine solution (200 mM diluted in $[D_6]$ DMSO) shared the following parameters: $SW_2=9.60$ ppm, $T_a=100$ μ s, $N=800$, $G_z=29.30$ Gcm⁻¹, and $G_{sz}=-5.84$ Gcm⁻¹ with a duration of 0.62 ms. Other experimental parameters for Z-PHASICS-SECSY were set as:

$SW_1=8.4$ ppm, and $n_i=512$ with a corresponding total acquisition duration of about 37 min. The experimental parameters of Z-PHASICS- J were set as: $SW_1=0.1$ ppm, and $n_i=64$, with corresponding total acquisition durations of about 6 min.

Third, the Z-PHASICS-HSQC and conventional HSQC experiments shared the following parameters: $SW_1=36.0$ ppm, $TR=5$ s, $ss=32$, $n_i=256$, the total acquisition durations of about 1 h and 25 min with four repetitions. Other experimental parameters of Z-PHASICS-HSQC were set as: $SW_2=6.7$ ppm, $T_d=143$ μ s, $N=800$, $G_z=29.30$ G cm $^{-1}$, and $G_{sz}=-5.84$ G cm $^{-1}$ with a duration of 0.4 ms. SW_2 of the conventional HSQC was 7.0 ppm.

Experiments utilizing R-PHASICS were carried out on a mixture comprising of ethyl 3-bromopropionate and 2-butanone in identical volumes to certify the robustness of R-PHASICS against inhomogeneity. The shimming coils of X1, Y1, and Z1 were detuned to introduce a 3D inhomogeneous field (denoted as XYZ inhomogeneity). According to the acquired field map, the magnetic field inhomogeneity was 0.034, 0.038, and 0.097 G cm $^{-1}$ along the X, Y, and Z axes, respectively. Experimental parameters shared by COSY and R-PHASICS-COSY were: $SW_1=5.00$ ppm, $SW_2=6.00$ ppm, and $n_i=256$ with corresponding total acquisition durations of about 17 min. The J -resolved and R-PHASICS- J experiments were performed with: $SW_1=0.20$ ppm, $SW_2=6.00$ ppm, and $n_i=50$ with corresponding acquisition durations of about 4 min. The R-PHASICS- J and R-PHASICS-COSY experiments shared the following parameters: $T_d=170$ μ s, $N=200$, $G_x=10.26$ G cm $^{-1}$, $G_y=11.35$ G cm $^{-1}$, $G_z=29.30$ G cm $^{-1}$, and $G_{sz}=-5.84$ G cm $^{-1}$ with the duration of 0.70 ms.

Quantitative analyses of resolution and SNR were carried out on spectra acquired by CONV and Z-PHASICS under a series of inhomogeneous fields induced by detuning the Z1 shimming coil. The used sample was a 19.0 mm ethyl 3-bromopropionate diluted in CDCl $_3$. Experiments under identical conditions were repeated five times every two minutes to ensure the fidelity of the measurements. Experimental parameters for Z-PHASICS were: $T_d=124$ μ s, $G_z=29.30$ G cm $^{-1}$, $SW_2=8.00$ ppm, $N=200$, and $G_{sz}=-5.84$ G cm $^{-1}$ with a duration of 0.70 ms.

More details in setting the experimental parameters are provided in the Supporting Information.

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